
SUMMARY

Senior Data Scientist with **5+ years of experience** in data science and ML, currently building GenAI evaluation systems. Recent focus spans golden dataset curation, LLM-as-judge frameworks, inter-annotator calibration, simulation-based adversarial testing, red-teaming protocols.

PROFESSIONAL EXPERIENCE

Senior Data Scientist, AI Center of Excellence, Fidelity Investments

April 2022 - Present

- Built the end-to-end evaluation system for a Representative Co-Pilot: annotation protocols, golden dataset curation, LLM-as-judge rubrics for extraction and summarization, judge-human validation via inter-annotator agreement, human-in-the-loop grading, and acceptance-criteria-based production drift monitoring through sampling; system serves **1K+** users, reducing workload by **40%**.
- Designed evaluation harnesses for a multi-step agentic workflow (information extraction $\rightarrow$ summarization), building LLM-as-judge suites that independently score each stage for accuracy, completeness, and faithfulness before passing deployment gates.
- Built simulation-based adversarial testing for a customer-facing chatbot: codified compliance rules as test oracles, iteratively probed the system, augmented attack vectors from failure logs to exploit discovered weaknesses, patched vulnerabilities, and ran A/B experiments on pre-patch vs. post-patch systems to quantify safety and reliability gains.
- Designed red-teaming protocols and layered safety evaluators using synthetic data generation, adversarial challenge sets, and structured error analysis, achieving **88%** F1 across **15** safety dimensions including toxicity and out-of-policy behavior.
- Architected and productionized a RAG system for **100K** call transcripts per day; built hallucination detection via fine-tuned NLI models, theme quantification, and hybrid retrieval quality evaluation, supporting **\$34M** in business insights for a **100+** person analytics team.
- Ran controlled prompt, model, and retrieval experiments measuring response quality, cost, latency, and production readiness across GenAI workflows; trained GPT-2 from scratch on clickstream data, improving session intent accuracy by **48%**.

Data Science Associate, ZS Associates

August 2020 - April 2022

- Delivered ML analytics, explainable AI, scenario analysis, and customer targeting for multiple pharma clients, including Fortune 500 companies, reducing sales effort by **20-30%**.
- Built mixed-integer optimization workflows for speaker programs while satisfying thousands of legal and business constraints.

SELECTED PROJECTS

- **CoexistAI – Agentic Research Framework**: Built a FastAPI- and LangChain-based agentic framework with multi-step tool use across web, PDF, code, and multimodal retrieval, pluggable LLM backends (OpenAI, hosted, local), and reproducible workflow orchestration.
- **smallGPT – From-Scratch LLM Systems**: Built GPT-style transformer systems from scratch in PyTorch, including KV caching, Flash Attention, RoPE, speculative decoding, and efficient inference pipelines.
- **modded-nanogpt Track 3 Optimization**: In an OpenAI researcher-hosted competition, set a step-count record (**3,125** steps to target loss) by extending SOAP-Muon with attention preconditioning and trust gating to improve LLM training step efficiency.

EDUCATION

Indian Institute of Technology Madras – MS by Research in Engineering Design (8.35/10.0) *July 2017 - July 2020*

Thesis: Interpretable Self Organizing Maps for Design Space Exploration and Optimization

Courses: Machine Learning for Engineering and Sciences, Applied Statistics, Engineering Optimization

Vishwakarma Institute of Technology – B.Tech. in Mechanical Engineering (8.81/10.0)

July 2013 - July 2017

PUBLICATIONS & RESEARCH

- Published IIT Madras research on self-organizing maps for interpretable multivariate optimization, reducing data requirements by $\sim 40\%$; co-authored WCSMO 13 paper on identifying robust design regions resilient to input uncertainty.

TECHNICAL STRENGTHS

AI Evals	Golden datasets, annotation protocols, LLM-as-judge, judge calibration, inter-annotator agreement, simulation-based adversarial testing, red teaming, deployment gates, drift monitoring.
GenAI	RAG systems, agentic workflows, prompt engineering, guardrails, hallucination detection, fine-tuning, NLI, chatbot architecture.
ML/Stats	Experimental design, NLP, scikit-learn, PyTorch, Optuna, NumPy, SciPy, pandas.
Backend/Infra	Python, SQL/MySQL, FastAPI, Docker, AWS, Git, async pipelines, vector retrieval, production ML workflows.